```python
from collections import deque
start = (3, 3, 1)
goal = (0, 0, 0)
moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]
queue = deque([(start, [])])
visited = {start}
while queue:
    state, path = queue.popleft()
    m, c, boat = state

    if state == goal:
        for step in path + [state]:
            print(step)
        break
    for dm, dc in moves:
        if boat == 1:
            new = (m - dm, c - dc, 0)
        else:
            new = (m + dm, c + dc, 1)
        nm, nc, nb = new
        mr, cr = 3 - nm, 3 - nc
        if 0 <= nm <= 3 and 0 <= nc <= 3:
            if (nm == 0 or nm >= nc) and (mr == 0 or mr >= cr):
                if new not in visited:
                    visited.add(new)
                    queue.append((new, path + [state]))
```

```
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)

Process finished with exit code 0
```